ASSESSMENT REMIT

Name			
ID#			
Semester	Fall 2026	**Section**	
School	School of Engineering, Architecture and Interior Design (EAID)		
Program	Bachelor of Science in Information Technology (BS IT)		
Major	Information Technology		
Course Code	**ITEC 300**		
Course Title	Computer Communications and Networking		
Level	☐ Remedial ☐ Bridging ☑ Undergraduate ☐ Graduate		
Assessment Title	Case Study — AI-Assisted Networking Application Development		
Weighting	20 Marks		
Instructor	Dr. Maher Alrahhal		
Hand Out Date	02/11/2026		
Due Date & Time	Date:14/12/2026	Time: 10:00 – 11:50 AM	
Feedback Post Date	14/12/2026		
Assessment Mode	☐ Individual ☑ Group ☐Closed Book ☑ Open Book		
Assignment Length *(for assignment-based assessments)*	Five-page report + working application		
Submission Mode	☐ Online by Email ☑ Online through LMS | TurnItIn ☐ Hard Copy		
Document Type	☐ Words ☑ PDF ☐ Artwork ☑ Software ☐ Other		

GENERAL REQUIREMENTS (CASE STUDY)
Students must work in groups of 3–4 members.
Each group must select one networking-related problem and develop a simple working website or application.
Students must use Visual Studio Code (VS Code) and may use an AI coding assistant such as OpenAI Codex, Claude, or Google Antigravity.
The application must demonstrate at least one correct networking concept, mechanism, service, or protocol.
Each group must submit a GitHub repository containing the complete source code and a README file.
Each group must submit a five-page PDF report.
Each group must demonstrate the working application and participate in a technical discussion.
All group members must understand the project and explain their individual contributions.
AI tools may assist with coding, but students are responsible for testing, understanding, and explaining their work.
Academic integrity and university assessment policies must be followed.

ASSESSMENT:

Case Study Scenario

An organization or university experiences networking challenges related to connectivity, performance, security, or network management.

Your group has been assigned to investigate one networking problem and develop a simple software application that demonstrates, analyzes, or helps address that problem.

Using VS Code and an AI coding assistant, develop a working website or application that applies concepts from Computer Communications and Networking.

Physical networking equipment is not required. Simulated networking scenarios are acceptable.

Suggested Case Study Topics

Project	Description
1. Smart Network Designer	Recommend network devices and generate a simple network topology based on user requirements.
2. Network Troubleshooting Assistant	Identify possible networking problems and recommend troubleshooting steps.
3. Network Monitoring Dashboard	Display simulated bandwidth, network traffic, and connection status.
4. LAN Simulator	Demonstrate communication between computers and switches in a LAN.
5. Routing Path Simulator	Show packet routes and how communication changes when a connection fails.
6. Network Security Dashboard	Demonstrate common security risks and protective mechanisms.
7. Smart Wi-Fi Network Planner	Recommend access points based on wireless users and coverage requirements.
8. WAN Design Assistant	Compare suitable WAN solutions for organizations with multiple locations.
9. Network Server Dashboard	Demonstrate DNS, DHCP, web server, and file server operations.

Groups may propose another networking-related application with instructor approval.

Only one main working networking functionality is required.

Question 1 — Networking Problem Identification and Analysis (4 Marks)

a) Identify the networking problem your application addresses. (1 Mark)

b) Explain why this problem is important for an organization or university. (1 Mark)

c) Identify two possible causes or challenges related to the problem. (1 Mark)

d) Explain how your proposed application will help analyze, demonstrate, or address the problem. (1 Mark)

Question 2 — Networking Concepts, Devices, and Protocols (6 Marks)

a) Identify and explain two networking concepts, mechanisms, services, or protocols related to your application. (2 Marks)

b) Identify the relevant networking devices or infrastructure components and explain their functions. (2 Marks)

c) Explain how these networking concepts are applied in your working application, using a practical example. (2 Marks)

Question 3 — Application Development and GitHub (5 Marks)

a) Develop a working website or application using VS Code and an AI coding assistant. The application must demonstrate at least one networking functionality. (3 Marks)

b) Upload the source code to GitHub and include a README file explaining how to run the application. (1 Mark)

c) Explain how the group used AI coding tools and improved the application during development. (1 Mark)

Question 4 — Testing, Evaluation, and Report (3 Marks)

a) Test the main functionality of your application and present the results. (1 Mark)

b) Evaluate the application by explaining one strength, one limitation, and one possible improvement. (1 Mark)

c) Submit a clear five-page PDF report documenting the project. (1 Mark)

Question 5 — Teamwork and Live Demonstration (2 Marks)

a) Demonstrate the working application and answer questions about its networking functionality. (1 Mark)

b) Explain each group member's responsibilities and contributions. (1 Mark)

Required Project Report — Five Pages

Page	Required Content
Page 1	Introduction, networking problem, objectives, and proposed application
Page 2	Networking concepts, devices, protocols, and application design
Page 3	AI-assisted development, VS Code, tools, and actual prompts
Page 4	Application screenshots, functionality, testing, and results

Page 5	Evaluation, limitations, conclusion, teamwork, and GitHub link

Final Submission

Each group must provide:

1. GitHub Repository: Complete application source code and README.
2. Five-Page PDF Report: Documentation, screenshots, AI prompts, testing, and evaluation.
3. Live Demonstration: Working application and technical discussion with the instructor.

ANSWER GUIDELINES:

NSWER GUIDELINES:

Students are expected to:

- Complete all five questions (Q1–Q5).
- Develop one working networking application.
- Explain the networking problem, relevant devices, protocols, and services.
- Demonstrate the application using realistic or simulated networking examples.
- Test the application and explain the results.
- Publish the complete source code on GitHub.
- Prepare a five-page PDF report.
- Participate in the live demonstration and discussion.

Presentation Requirements:

- Include all student names and IDs.
- Clearly identify the selected case study topic.
- Include screenshots of the working application.
- Include actual AI prompts used during development.
- Provide the GitHub repository link.
- Explain each group member's contribution.
- Use clear technical explanations and appropriate references.

REFERENCING STYLE:

Students should use the American Psychological Association style (APA) as per the following guidelines: https://apastyle.apa.org/style-grammar-guidelines/references/examples

PLAGIARISM & GENERATIVE AI (FOR ASSIGNMENT BASED ASSESSMENTS):

Amity University Dubai is committed to academic honesty and integrity.
For this case study, students are permitted to use AI coding assistants, including OpenAI Codex, Claude, and Google Antigravity, to support application development.
Students must:

- Document the AI tools and actual prompts used.
- Review and test all AI-generated code.
- Understand and explain the application and its networking functionality.
- Write their own analysis, evaluation, and conclusions.
- Acknowledge external sources and AI assistance appropriately.

Submitting work that students cannot explain or presenting AI-generated material as entirely independent work may constitute academic misconduct under university policy.

LATE SUBMISSIONS (FOR ASSIGNMENT BASED ASSESSMENTS):

You are advised to strictly adhere to the deadline. While Amity University Dubai might consider extenuating circumstances to support students who are experiencing unforeseen issues impacting their ability to complete assessments on time, any anticipated submission delay must be communicated in advance to the instructor who will consult with the Program Leader and the Dean before a permission is granted. All permissions must be based on valid reasons backed up by concrete evidence. For more details, you are advised to consult the latest version of AMUD's Academic Integrity and Academic Misconduct Policy.

FEEDBACK PLAN:

The preparation for tasks included in this assessment will be supported by formative feedback during the class regular sessions and related office hours. Summative written feedback will be provided on the date indicated earlier.

ASSESSED COURSE LEARNING OUTCOMES:	RELATED QUESTIONS	MAX SCORE	MARK
CLO 3: Apply principles of computing to identify the basic mechanisms, services and protocols for the organizational network infrastructure.	Q1–Q2	**10**	
CLO 4: Work in a team to evaluate the devices and protocols to design a network using suitable explanations for the given problem.	Q3–Q5	**10**	
	TOTAL	**20**	

EMBEDDED PLO DIRECT ASSESSMENT QUESTIONS? ☑ **YES** ☐ **NO**
[IF YES, COMPLETE THE TABLE BELOW]

PLO 1	EMBEDDED QUESTIONS

Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	CLO 3: Q1–Q2 (10 marks). CLO 4: Q3–Q5 (10 marks). Total: 20 marks.

SCORING RUBRIC			LEVEL OF ACHIEVEMENT			
	DESCRIPTOR	**WEIGHT**	**4 [87%-100%]**	**3 [73%-86%]**	**2 [60%-72%]**	**1 [0%-59%]**
1	CLO 3: Networking Problem Analysis (Q1)	4	Clearly identifies and analyzes the problem, causes, and proposed solution.	Mostly correct analysis with minor omissions.	Partial analysis with limited explanations.	Incorrect or missing problem analysis.
2	CLO 3: Networking Concepts and Protocols (Q2)	6	Accurately explains and applies relevant concepts, devices, and protocols.	Mostly correct explanations and applications.	Partial understanding with several errors.	Incorrect or missing networking concepts.
3	CLO 4: Application Development and GitHub (Q3)	5	Fully functional application, correct networking functionality, and complete GitHub repository.	Mostly functional with minor issues.	Partially functional with incomplete documentation.	Nonfunctional or missing application.
4	CLO 4: Testing, Evaluation, and Report (Q4)	3	Clear testing, meaningful evaluation, and complete five-page report.	Mostly complete with minor omissions.	Limited testing or incomplete evaluation.	Missing or incorrect testing, evaluation, or report.
5	CLO 4: Teamwork and Demonstration (Q5)	2	All members contribute and clearly demonstrate and explain the application.	Most members contribute and explain the work.	Limited participation or incomplete demonstration.	Minimal teamwork or unsuccessful demonstration.

Student Declaration

By signing this cover page, you acknowledge the instructions above (general requirements) and accept all responsibilities resulting from breaching them as per the related University policies. I hereby declare that it is my original work and I have duly acknowledged and referenced the source of any quotes, thoughts or ideas incorporated in the work.

Student Signature _____ **Date** _____

To be filled by faculty

FEEDBACK FROM FACULTY:

Faculty Name and Signature **Date**